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Lem convergencia lp imp medida

Lema 1. *La convergencia en $\mathcal{L}^p$ implica la convergencia en medida.*

***Demostración*:** Sea (X, Σ, μ) un espacio de medida y $(f_n)_{n\in\mathbb{N}} \subset \mathcal{L}^p$ tal que $f_n \xrightarrow[n\to\infty]{\mathcal{L}^p} f$. Sea $\lambda > 0$, definimos $\forall n \in \mathbb{N} : A_n := \{x \in X : |f_n(x) - f(x)| > \lambda\}$. Entonces,

$$0 \leqslant \mu(A_n) = \int_{A_n} 1 \,\mathrm{d}\mu(x) \leqslant \int_X \frac{|f_n(x) - f(x)|^p}{\lambda^p} \,\mathrm{d}\mu(x) = \frac{1}{\lambda^p} \|f_n - f\|_p^p .$$

Ahora, como $\lim_{n\to\infty} \|f_n - f\|_p = 0$, tenemos que $\lim_{n\to\infty} \mu(A_n) = 0$. ■

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